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INSTITUT UNIVERSITAIRE DES SCIENCES (IUS)

Faculté des sciences et des technologies (FST)

Administration Système TD_8

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Niveau : L3

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L'objectif de ce TD est de :

1. Mettre en place une solution complète de supervision d'un serveur Ubuntu :
 - collecte des métriques systèmes
 - visualisation graphique
 - détection d'incidents
 - alertes

❖ Installation de Prometheus

```
bribri@bribri1: ~  
1 additional security update can be applied with ESM Apps.  
Learn more about enabling ESM Apps service at https://ubuntu.com/esm  
  
Last login: Sun Apr 19 18:49:26 2026 from 192.168.1.128  
bribri@bribri1:~$ sudo apt update  
[sudo] password for bribri:  
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease  
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease  
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease  
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
44 packages can be upgraded. Run 'apt list --upgradable' to see them.  
bribri@bribri1:~$ sudo apt upgrade -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
Calculating upgrade... Done  
Get another security update through Ubuntu Pro with 'esm-apps' enabled:  
  liburiparser1  
Learn more about Ubuntu Pro at https://ubuntu.com/pro  
The following NEW packages will be installed:  
  linux-headers-6.8.0-110 linux-headers-6.8.0-110-generic  
  linux-image-6.8.0-110-generic linux-modules-6.8.0-110-generic  
  linux-modules-extra-6.8.0-110-generic linux-tools-6.8.0-110  
  linux-tools-6.8.0-110-generic  
The following packages will be upgraded:  
  apparmor gir1.2-packagekitglib-1.0 jq libapparmor1 libcap2 libcap2-bin  
  libjq1 libnss-systemd libntfs-3g89t64 libpackagekit-glib2-18 libpam-cap  
  libpam-systemd libpolkit-agent-1-0 libpolkit-gobject-1-0  
  libpython3.12-minimal libpython3.12-stdlib libpython3.12t64  
  libsystemd-shared libsystemd0 libudev1 linux-firmware linux-generic  
  linux-headers-generic linux-image-generic linux-libc-dev  
  linux-tools-common lshw ntfs-3g packagekit packagekit-tools polkitd  
  python3.12 python3.12-minimal rsyslog snapd systemd systemd-dev  
  systemd-resolved systemd-sysv systemd-timesyncd ubuntu-drivers-common  
  ubuntu-pro-client ubuntu-pro-client-l10n udev  
44 upgraded, 7 newly installed, 0 to remove and 0 not upgraded.  
25 standard LTS security updates
```

➤ Creation de l'utilisateur systeme

```
bribri@bribri1:~$ sudo useradd --no-create-home --shell /bin/false prometheus  
bribri@bribri1:~$
```

➤ Creation des dossiers

```
bribri@bribri1:~$ sudo mkdir /etc/prometheus
bribri@bribri1:~$ sudo mkdir /var/lib/prometheus
bribri@bribri1:~$
```

➤ Telechargement de prometheus

```
bribri@bribri1:~$ wget https://github.com/prometheus/prometheus/releases/download/v2.45.0/prometheus-2.45.0.linux-amd64.tar.gz
--2026-04-27 03:07:12-- https://github.com/prometheus/prometheus/releases/download/v2.45.0/prometheus-2.45.0.linux-amd64.tar.gz
Resolving github.com (github.com)... 140.82.113.3
Connecting to github.com (github.com)|140.82.113.3|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://release-assets.githubusercontent.com/github-production-release-asset/6838921/75753d2b-6c1e-4ac8-bdee-941c26d6aa46?sp=r&sv=2018-11-09&sr=b&spr=https&se=2026-04-27T03%3A58%3A55Z&rsct=attachment%3B+filename%3Dprometheus-2.45.0.linux-amd64.tar.gz&rsct=application%2Foctet-stream&skoid=96c2d410-5711-43a1-aedd-ab1947aa7ab0&sktid=398a6654-997b-47e9-b12b-9515b896b4de&skt=2026-04-27T02%3A58%3A33Z&ske=2026-04-27T03%3A58%3A55Z&sks=b&skv=2018-11-09&sig=54kMWL5e3gP00wo16os7Qrv9s8krPp6PU9zDp0X9oj0%3D&jwt=eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJpc3MiOiJnaXRodWIuY29tIiwiaXVkiJjoicmVsZWZzZS1hc3NldHMmuZ2l0aHViZXRlcmlmNvbnRlbnQuY29tIiwia2V5Ijoia2V5MSIsImV4cCI6MTc3NzI2MTAzMywibmJmIjoxNzc3MjU5MjMzLCJwYXRoIjoicmVsZWZzZWZzZ2V0cHJvZHVjdGlubi5ibG9iLmNvcmlud2luZG93cy5uZXQifQ.vS8xEWu1xmOoHgG2R-YK65cwrTSdUfn-oagL7SSYIMQ&response-content-disposition=attachment%3B%20filename%3Dprometheus-2.45.0.linux-amd64.tar.gz&response-content-type=application%2Foctet-stream [following]
--2026-04-27 03:07:13-- https://release-assets.githubusercontent.com/github-production-release-asset/6838921/75753d2b-6c1e-4ac8-bdee-941c26d6aa46?sp=r&sv=2018-11-09&sr=b&spr=https&se=2026-04-27T03%3A58%3A55Z&rsct=attachment%3B+filename%3Dprometheus-2.45.0.linux-amd64.tar.gz&rsct=application%2Foctet-stream&skoid=96c2d410-5711-43a1-aedd-ab1947aa7ab0&sktid=398a6654-997b-47e9-b12b-9515b896b4de&skt=2026-04-27T02%3A58%3A33Z&ske=2026-04-27T03%3A58%3A55Z&sks=b&skv=2018-11-09&sig=54kMWL5e3gP00wo16os7Qrv9s8krPp6PU9zDp0X9oj0%3D&jwt=eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJpc3MiOiJnaXRodWIuY29tIiwiaXVkiJjoicmVsZWZzZS1hc3NldHMmuZ2l0aHViZXRlcmlmNvbnRlbnQuY29tIiwia2V5Ijoia2V5MSIsImV4cCI6MTc3NzI2MTAzMywibmJmIjoxNzc3MjU5MjMzLCJwYXRoIjoicmVsZWZzZWZzZ2V0cHJvZHVjdGlubi5ibG9iLmNvcmlud2luZG93cy5uZXQifQ.vS8xEWu1xmOoHgG2R-YK65cwrTSdUfn-oagL7SSYIMQ&response-content-disposition=attachment%3B%20filename%3Dprometheus-2.45.0.linux-amd64.tar.gz&response-content-type=application%2Foctet-stream
Resolving release-assets.githubusercontent.com (release-assets.githubusercontent.com)... 185.199.110.133, 185.199.109.133, 185.199.111.133, ...
```

```
bribri@bribri1:~$ tar -xvzf prometheus-2.45.0.linux-amd64.tar.gz
prometheus-2.45.0.linux-amd64/
prometheus-2.45.0.linux-amd64/LICENSE
prometheus-2.45.0.linux-amd64/prometheus.yml
prometheus-2.45.0.linux-amd64/console_libraries/
prometheus-2.45.0.linux-amd64/console_libraries/prom.lib
prometheus-2.45.0.linux-amd64/console_libraries/menu.lib
prometheus-2.45.0.linux-amd64/consoles/
prometheus-2.45.0.linux-amd64/consoles/node-overview.html
prometheus-2.45.0.linux-amd64/consoles/node-cpu.html
prometheus-2.45.0.linux-amd64/consoles/index.html.example
prometheus-2.45.0.linux-amd64/consoles/node.html
prometheus-2.45.0.linux-amd64/consoles/node-disk.html
prometheus-2.45.0.linux-amd64/consoles/prometheus-overview.html
prometheus-2.45.0.linux-amd64/consoles/prometheus.html
prometheus-2.45.0.linux-amd64/promtool
prometheus-2.45.0.linux-amd64/prometheus

gzip: stdin: unexpected end of file
tar: Unexpected EOF in archive
tar: Unexpected EOF in archive
tar: Error is not recoverable: exiting now
bribri@bribri1:~$ tar -xvzf prometheus-2.45.0.linux-amd64.tar.gz
```

➤ Copier les fichiers

```
bribri@bribri1:~$ sudo mkdir /etc/prometheus
mkdir: cannot create directory '/etc/prometheus': File exists
bribri@bribri1:~$ cd prometheus-2.45.0.linux-amd64
bribri@bribri1:~/prometheus-2.45.0.linux-amd64$ sudo cp prometheus /usr/local/bin/
bribri@bribri1:~/prometheus-2.45.0.linux-amd64$ sudo cp promtool /usr/local/bin/
bribri@bribri1:~/prometheus-2.45.0.linux-amd64$ sudo cp -r consoles /etc/prometheus/
bribri@bribri1:~/prometheus-2.45.0.linux-amd64$
```

➤ Configuration principale

```
bribri@bribri1:~/prometheus-2.45.0.linux-amd64$ sudo cp -r consoles /etc/prometheus/
bribri@bribri1:~/prometheus-2.45.0.linux-amd64$ sudo nano /etc/prometheus/prometheus.yml
bribri@bribri1:~/prometheus-2.45.0.linux-amd64$ sudo cat /etc/systemd/system/prometheus.service
cat: /etc/systemd/system/prometheus.service: No such file or directory
bribri@bribri1:~/prometheus-2.45.0.linux-amd64$ sudo cat /etc/prometheus/prometheus.yml
global:
  scrape_interval: 15s

scrape_configs:
  - job_name: 'prometheus'
    static_configs:
      - targets: ['localhost:9090']

  - job_name: 'node_exporter'
    static_configs:
      - targets: ['localhost:9100']

  - job_name: 'nginx'
    static_configs:
      - targets: ['localhost:9113']
bribri@bribri1:~/prometheus-2.45.0.linux-amd64$
```

➤ Service systemd

Creation:

```
bribri@bribri1:~/prometheus-2.45.0.linux-amd64$ sudo cat /etc/prometheus/prometheus.yml
global:
  scrape_interval: 15s

scrape_configs:
  - job_name: 'prometheus'
    static_configs:
      - targets: ['localhost:9090']

  - job_name: 'node_exporter'
    static_configs:
      - targets: ['localhost:9100']

  - job_name: 'nginx'
    static_configs:
      - targets: ['localhost:9113']
bribri@bribri1:~/prometheus-2.45.0.linux-amd64$
```

➤ Donner les permissions

```
pribri@bribril:~/prometheus-2.45.0.linux-amd64$ sudo chown -R prometheus:prometheus /etc/prometheus
pribri@bribril:~/prometheus-2.45.0.linux-amd64$ sudo chown -R prometheus:prometheus /var/lib/prometheus
pribri@bribril:~/prometheus-2.45.0.linux-amd64$ sudo chmod -R 775 /var/lib/prometheus
pribri@bribril:~/prometheus-2.45.0.linux-amd64$
```

➤ **Demarrage**

```

bribr1@bribr1:~/.prometheus-2.45.0.linux-amd64$ sudo systemctl daemon-reload
bribr1@bribr1:~/.prometheus-2.45.0.linux-amd64$ sudo systemctl start prometheus
bribr1@bribr1:~/.prometheus-2.45.0.linux-amd64$ sudo systemctl enable prometheus
Created symlink /etc/systemd/system/multi-user.target.wants/prometheus.service → /etc/systemd/system/prometheus.service.
bribr1@bribr1:~/.prometheus-2.45.0.linux-amd64$ sudo systemctl status prometheus

```

➤ Verification

```

wribri@bribril:~/prometheus-2.45.0-linux-amd64$ sudo systemctl status prometheus
* prometheus.service - Prometheus
   Loaded: loaded (/etc/systemd/system/prometheus.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-04-27 04:11:51 UTC; 2min 42s ago
   Main PID: 2609 (prometheus)
     Tasks: 8 (limit: 4608)
    Memory: 20.7M (peak: 21.1M)
       CPU: 1.443s
   CGroup: /system.slice/prometheus.service
           └─2609 /usr/local/bin/prometheus --config.file=/etc/prometheus/prometheus.yml --storage.tsdb.path=/var/lib/prometheus/

Apr 27 04:11:51 bribril prometheus[2609]: ts=2026-04-27T04:11:51.602z caller=head.go:676 level=info component=tsdb msg="On-disk memory mappable chunks replay completed"
Apr 27 04:11:51 bribril prometheus[2609]: ts=2026-04-27T04:11:51.602z caller=head.go:684 level=info component=tsdb msg="Replaying WAL, this may take a while"
Apr 27 04:11:51 bribril prometheus[2609]: ts=2026-04-27T04:11:51.607z caller=head.go:755 level=info component=tsdb msg="WAL segment loaded" segment=0 maxSegment=0
Apr 27 04:11:51 bribril prometheus[2609]: ts=2026-04-27T04:11:51.607z caller=head.go:792 level=info component=tsdb msg="WAL replay completed" checkpoint_replay_duration=
Apr 27 04:11:51 bribril prometheus[2609]: ts=2026-04-27T04:11:51.617z caller=main.go:1048 level=info fs_type=EXT4_SUPER_MAGIC
Apr 27 04:11:51 bribril prometheus[2609]: ts=2026-04-27T04:11:51.617z caller=main.go:1043 level=info msg="TSDB started"
Apr 27 04:11:51 bribril prometheus[2609]: ts=2026-04-27T04:11:51.617z caller=main.go:1224 level=info msg="Loading configuration file" filename=/etc/prometheus/prometheus
Apr 27 04:11:51 bribril prometheus[2609]: ts=2026-04-27T04:11:51.873z caller=main.go:1261 level=info msg="Completed loading of configuration file" filename=/etc/promethe
Apr 27 04:11:51 bribril prometheus[2609]: ts=2026-04-27T04:11:51.874z caller=main.go:1044 level=info msg="Server is ready to receive web requests."
Apr 27 04:11:51 bribril prometheus[2609]: ts=2026-04-27T04:11:51.874z caller=manager.go:995 level=info component="rule manager" msg="Starting rule manager..."

[27]# Stopped
wribri@bribril:~/prometheus-2.45.0-linux-amd64$

```

➤ Installation de Node Exporter

[illegible]

```

bribri@bribri1: ~/node_expor  X  +  v
linux-amd64.tar.gz&response-content-type=application%2Foctet-stream
Resolving release-assets.githubusercontent.com (release-assets.githubusercontent.com)... 18
5.199.110.133, 185.199.109.133, 185.199.111.133, ...
Connecting to release-assets.githubusercontent.com (release-assets.githubusercontent.com)|1
85.199.110.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 10368103 (9.9M) [application/octet-stream]
Saving to: 'node_exporter-1.6.1.linux-amd64.tar.gz'

node_exporter-1.6. 100%[=====>] 9.89M 27.5KB/s in 6m 35s

2026-06-05 10:57:58 (25.6 KB/s) - 'node_exporter-1.6.1.linux-amd64.tar.gz' saved [10368103/
10368103]

node_exporter-1.6.1.linux-amd64/
node_exporter-1.6.1.linux-amd64/NOTICE
bribri@bribri1:~/node_exporter-1.6.1.linux-amd64$ sudo chown node_exporter:node_exporter /u
sr/local/bin/node_exporter
bribri@bribri1:~/node_exporter-1.6.1.linux-amd64$ sudo nano /etc/systemd/system/node_export
er.service
bribri@bribri1:~/node_exporter-1.6.1.linux-amd64$ sudo systemctl daemon-reload
bribri@bribri1:~/node_exporter-1.6.1.linux-amd64$ sudo systemctl start node_exporter
bribri@bribri1:~/node_exporter-1.6.1.linux-amd64$ sudo systemctl enable node_exporter
Created symlink /etc/systemd/system/multi-user.target.wants/node_exporter.service → /etc/sy
stemd/system/node_exporter.service.
bribri@bribri1:~/node_exporter-1.6.1.linux-amd64$ sudo systemctl status node_exporter
● node_exporter.service - Node Exporter
   Loaded: loaded (/etc/systemd/system/node_exporter.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-06-05 11:01:26 UTC; 24s ago
     Main PID: 46129 (node_exporter)
        Tasks: 5 (limit: 4601)
       Memory: 5.4M (peak: 5.6M)
          CPU: 152ms
      CGroup: /system.slice/node_exporter.service
              └─46129 /usr/local/bin/node_exporter

Jun 05 11:01:27 bribri1 node_exporter[46129]: ts=2026-06-05T11:01:27.033Z caller=node_expo>
Jun 05 11:01:27 bribri1 node_exporter[46129]: ts=2026-06-05T11:01:27.033Z caller=node_expo>
Jun 05 11:01:27 bribri1 node_exporter[46129]: ts=2026-06-05T11:01:27.033Z caller=node_expo>
Jun 05 11:01:27 bribri1 node_exporter[46129]: ts=2026-06-05T11:01:27.033Z caller=node_expo>
Jun 05 11:01:27 bribri1 node_exporter[46129]: ts=2026-06-05T11:01:27.033Z caller=node_expo>
Jun 05 11:01:27 bribri1 node_exporter[46129]: ts=2026-06-05T11:01:27.033Z caller=node_expo>
Jun 05 11:01:27 bribri1 node_exporter[46129]: ts=2026-06-05T11:01:27.033Z caller=node_expo>
Jun 05 11:01:27 bribri1 node_exporter[46129]: ts=2026-06-05T11:01:27.035Z caller=tlscnfi>
Jun 05 11:01:27 bribri1 node_exporter[46129]: ts=2026-06-05T11:01:27.036Z caller=tlscnfi>
Jun 05 11:01:36 bribri1 node_exporter[46129]: ts=2026-06-05T11:01:36.353Z caller=collecto>
lines 1-20/20 (END)
[1]+  Stopped                  sudo systemctl status node_exporter
bribri@bribri1:~/node_exporter-1.6.1.linux-amd64$

```


➤ Installation de Nginx + Nginx Exporter

```
bribri@bribri1: ~  
bribri@bribri1:~$ sudo apt install nginx -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following additional packages will be installed:  
  nginx-common  
Suggested packages:  
  fcgiwrap nginx-doc  
The following packages will be upgraded:  
  nginx nginx-common  
2 upgraded, 0 newly installed, 0 to remove and 110 not upgraded.  
Need to get 566 kB of archives.  
After this operation, 2,048 B of additional disk space will be used.  
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx amd64  
1.24.0-2ubuntu7.9 [522 kB]  
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx-common  
all 1.24.0-2ubuntu7.9 [44.2 kB]  
Fetched 566 kB in 4s (127 kB/s)  
apt-listchanges: Reading changelogs...  
Preconfiguring packages ...  
(Reading database ... 156027 files and directories currently installed.)  
Preparing to unpack .../nginx_1.24.0-2ubuntu7.9_amd64.deb ...  
Unpacking nginx (1.24.0-2ubuntu7.9) over (1.24.0-2ubuntu7.6) ...  
Preparing to unpack .../nginx-common_1.24.0-2ubuntu7.9_all.deb ...  
Unpacking nginx-common (1.24.0-2ubuntu7.9) over (1.24.0-2ubuntu7.6) ...  
Setting up nginx-common (1.24.0-2ubuntu7.9) ...  
Setting up nginx (1.24.0-2ubuntu7.9) ...  
  * Upgrading binary nginx  
    [ OK ]  
Processing triggers for man-db (2.12.0-4build2) ...  
Processing triggers for ufw (0.36.2-6) ...  
Scanning processes...  
Scanning linux images...  
  
Running kernel seems to be up-to-date.  
  
No services need to be restarted.  
  
No containers need to be restarted.  
  
No user sessions are running outdated binaries.
```

```
bribri@bribri1: ~  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
bribri@bribri1:~$ sudo systemctl enable --now nginx  
Synchronizing state of nginx.service with SysV service script with /usr/lib/  
systemd/systemd-sysv-install.  
bribri@bribri1:~$ sudo nano /etc/nginx/sites-available/default  
bribri@bribri1:~$ sudo nginx -t  
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok  
nginx: configuration file /etc/nginx/nginx.conf test is successful  
bribri@bribri1:~$ sudo systemctl restart nginx  
bribri@bribri1:~$ wget https://github.com/nginxinc/nginx-prometheus-exporter/  
releases/download/v0.11.0/nginx-prometheus-exporter_0.11.0_linux_amd64.tar.  
gz  
--2026-06-05 11:13:10-- https://github.com/nginxinc/nginx-prometheus-export  
er/releases/download/v0.11.0/nginx-prometheus-exporter_0.11.0_linux_amd64.ta  
r.gz  
Resolving github.com (github.com)... 140.82.113.4  
Connecting to github.com (github.com)|140.82.113.4|:443... connected.  
HTTP request sent, awaiting response... 301 Moved Permanently  
Location: https://github.com/nginx/nginx-prometheus-exporter/releases/downlo  
ad/v0.11.0/nginx-prometheus-exporter_0.11.0_linux_amd64.tar.gz [following]  
--2026-06-05 11:13:11-- https://github.com/nginx/nginx-prometheus-exporter/  
releases/download/v0.11.0/nginx-prometheus-exporter_0.11.0_linux_amd64.tar.g  
z  
Reusing existing connection to github.com:443.  
HTTP request sent, awaiting response... 302 Found  
Location: https://release-assets.githubusercontent.com/github-production-rel  
ease-asset/135449032/5b144f95-bc89-4dfe-a908-582ccf41ffef?sp=r&sv=2018-11-09
```

```
bribri@bribri1:~$ tar xvf nginx-prometheus-exporter_0.11.0_linux_amd64.tar.  
gz  
CHANGELOG.md  
LICENSE  
README.md  
nginx-prometheus-exporter  
  
bribri@bribri1:~$ sudo systemctl daemon-reload  
bribri@bribri1:~$ sudo systemctl start nginx_exporter  
bribri@bribri1:~$ sudo systemctl enable nginx_exporter  
Created symlink /etc/systemd/system/multi-user.target.wants/nginx_exporter.s  
ervice → /etc/systemd/system/nginx_exporter.service.  
bribri@bribri1:~$ sudo systemctl status nginx_exporter  
× nginx_exporter.service - Nginx Prometheus Exporter  
   Loaded: loaded (/etc/systemd/system/nginx_exporter.service; enabled; p>  
   Active: failed (Result: exit-code) since Fri 2026-06-05 11:19:47 UTC; >  
   Duration: 5.035s  
   Main PID: 47011 (code=exited, status=1/FAILURE)  
    CPU: 26ms  
  
Jun 05 11:19:42 bribri1 systemd[1]: Started nginx_exporter.service - Nginx >  
Jun 05 11:19:42 bribri1 nginx-prometheus-exporter[47011]: NGINX Prometheus >  
Jun 05 11:19:42 bribri1 nginx-prometheus-exporter[47011]: 2026/06/05 11:19: >  
Jun 05 11:19:47 bribri1 nginx-prometheus-exporter[47011]: 2026/06/05 11:19: >  
Jun 05 11:19:47 bribri1 systemd[1]: nginx_exporter.service: Main process ex >  
Jun 05 11:19:47 bribri1 systemd[1]: nginx_exporter.service: Failed with res >  
lines 1-13/13 (END)  
[3]+  Stopped                                sudo systemctl status nginx_exporter  
bribri@bribri1:~$
```

➤ Installation de Grafana

```
bribri@bribri1: ~  
E: Unable to fetch some archives, maybe run apt-get update or try with --fix-missing?  
bribri@bribri1:~$ sudo apt install grafana -y  
[sudo] password for bribri:  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following NEW packages will be installed:  
  grafana  
0 upgraded, 1 newly installed, 0 to remove and 110 not upgraded.  
Need to get 279 MB of archives.  
After this operation, 1,039 MB of additional disk space will be used.  
Get:1 https://packages.grafana.com/oss/deb stable/main amd64 grafana amd64 13.0.2 [279 MB]  
59% [1 grafana 205 MB/279 MB 74%] 13.6 kB/s 1h 31m Ign:1 https://packages.grafana  
.com/oss/deb stable/main amd64 grafana amd64 13.0.2  
Get:1 https://packages.grafana.com/oss/deb stable/main amd64 grafana amd64 13.0.2 [279 MB]  
61% [1 grafana 212 MB/279 MB 76%] 90.5 kB/s 12min 14s b76% [1 grafana 265 MB/279 MB 9  
5%] 136 Fetched 73.2 MB in 18min 48s (64.8 kB/s)  
Selecting previously unselected package grafana.  
(Reading database ... 156031 files and directories currently installed.)  
Preparing to unpack ../grafana_13.0.2_amd64.deb ...  
Unpacking grafana (13.0.2) ...  
Setting up grafana (13.0.2) ...  
info: Selecting UID from range 100 to 999 ...  
  
info: Adding system user `grafana' (UID 113) ...  
info: Adding new user `grafana' (UID 113) with group `grafana' ...  
info: Not creating home directory `/usr/share/grafana'.  
### NOT starting on installation, please execute the following statements to configure graf  
ana to start automatically using systemd  
  sudo /bin/systemctl daemon-reload  
  sudo /bin/systemctl enable grafana-server  
### You can start grafana-server by executing  
  sudo /bin/systemctl start grafana-server  
Scanning processes...  
Scanning linux images...  
  
Running kernel seems to be up-to-date.  
  
No services need to be restarted.  
  
No containers need to be restarted.  
  
No user sessions are running outdated binaries.  
  
No VM guests are running outdated hypervisor (qemu)  
  binaries on this host.  
bribri@bribri1:~$ sudo systemctl enable --now grafana-server  
[sudo] password for bribri:  
Synchronizing state of grafana-server.service with SysV service script with /usr/lib/system
```

```
bribri@bribri1: ~  
No services need to be restarted.  
No containers need to be restarted.  
No user sessions are running outdated binaries.  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
bribri@bribri1:~$ sudo systemctl enable --now grafana-server  
[sudo] password for bribri:  
Synchronizing state of grafana-server.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.  
Executing: /usr/lib/systemd/systemd-sysv-install enable grafana-server  
Created symlink /etc/systemd/system/multi-user.target.wants/grafana-server.service → /usr/lib/systemd/system/grafana-server.service.  
bribri@bribri1:~$ sudo systemctl status grafana-server  
● grafana-server.service - Grafana instance  
   Loaded: loaded (/usr/lib/systemd/system/grafana-server.service; vendor preset: enabled)  
   Active: active (running) since Fri 2026-06-05 20:48:11 CEST; 1min 22s ago  
     Docs: http://docs.grafana.org  
    Main PID: 49887 (grafana)  
      Tasks: 7 (limit: 4601)  
    Memory: 42.2M (peak: 42.5M)  
       CPU: 6.397s  
    CGroup: /system.slice/grafana-server.service  
            └─49887 /usr/share/grafana/bin/grafana s>  
  
Jun 05 20:48:11 bribri1 grafana[49887]: logger=migrat>  
Jun 05 20:48:11 bribri1 grafana[49887]: logger=migrat>  
Jun 05 20:48:11 bribri1 grafana[49887]: logger=migrat>  
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Jun 05 20:48:11 bribri1 grafana[49887]: logger=migrat>  
Jun 05 20:48:11 bribri1 grafana[49887]: logger=migrat>  
lines 1-22/22 (END)  
[2]+  Stopped                  sudo systemctl status grafana-server  
bribri@bribri1:~$
```

➤ Vérification finale de tous les services

```
bribri@bribri1: ~  
  
Loaded: loaded (/etc/systemd/system/prometheus.service; enabled; preset: enabled)  
Active: active (running) since Mon 2026-04-27 04:11:51 UTC; 1 month 9 days ago  
Main PID: 2609 (prometheus)  
Tasks: 8 (limit: 4601)  
Memory: 37.5M (peak: 64.5M)  
CPU: 54.044s  
CGroup: /system.slice/prometheus.service  
└─2609 /usr/local/bin/prometheus --config.file=/etc/prometheus/prometheus.yml --storage.tsdb.path=/v  
  
Apr 27 04:11:51 bribri1 prometheus[2609]: ts=2026-04-27T04:11:51.874Z caller=manager.go:995 level=info component=>  
Jun 05 10:45:48 bribri1 prometheus[2609]: ts=2026-06-05T10:45:48.084Z caller=compact.go:523 level=info component=>  
Jun 05 10:45:48 bribri1 prometheus[2609]: ts=2026-06-05T10:45:48.056Z caller=head.go:1293 level=info component=ts>  
Jun 05 18:32:12 bribri1 prometheus[2609]: ts=2026-06-05T18:32:12.810Z caller=compact.go:514 level=info component=>  
Jun 05 18:32:12 bribri1 prometheus[2609]: ts=2026-06-05T18:32:12.828Z caller=head.go:1293 level=info component=ts>  
Jun 05 18:32:13 bribri1 prometheus[2609]: ts=2026-06-05T18:32:13.165Z caller=compact.go:523 level=info component=>  
Jun 05 18:32:13 bribri1 prometheus[2609]: ts=2026-06-05T18:32:13.181Z caller=db.go:1617 level=info component=tsdb>  
Jun 05 18:32:13 bribri1 prometheus[2609]: ts=2026-06-05T18:32:13.185Z caller=head.go:1293 level=info component=ts>  
Jun 05 20:32:22 bribri1 prometheus[2609]: ts=2026-06-05T20:32:22.662Z caller=compact.go:514 level=info component=>  
Jun 05 20:32:22 bribri1 prometheus[2609]: ts=2026-06-05T20:32:22.666Z caller=head.go:1293 level=info component=ts>  
lines 1-20/20 (END)  
[3]+ Stopped sudo systemctl status prometheus  
bribri@bribri1:~$ sudo systemctl status node_exporter  
● node_exporter.service - Node Exporter  
Loaded: loaded (/etc/systemd/system/node_exporter.service; enabled; preset: enabled)  
Active: active (running) since Fri 2026-06-05 11:01:26 UTC; 9h ago  
Main PID: 46129 (node_exporter)  
Tasks: 5 (limit: 4601)  
Memory: 8.3M (peak: 9.2M)  
CPU: 41.337s  
CGroup: /system.slice/node_exporter.service  
└─46129 /usr/local/bin/node_exporter  
  
Jun 05 20:50:11 bribri1 node_exporter[46129]: ts=2026-06-05T20:50:11.192Z caller=collector.go:169 level=error msg>  
Jun 05 20:50:26 bribri1 node_exporter[46129]: ts=2026-06-05T20:50:26.192Z caller=collector.go:169 level=error msg>  
Jun 05 20:50:41 bribri1 node_exporter[46129]: ts=2026-06-05T20:50:41.176Z caller=collector.go:169 level=error msg>  
Jun 05 20:50:56 bribri1 node_exporter[46129]: ts=2026-06-05T20:50:56.228Z caller=collector.go:169 level=error msg>  
Jun 05 20:51:11 bribri1 node_exporter[46129]: ts=2026-06-05T20:51:11.234Z caller=collector.go:169 level=error msg>  
Jun 05 20:51:26 bribri1 node_exporter[46129]: ts=2026-06-05T20:51:26.202Z caller=collector.go:169 level=error msg>  
Jun 05 20:51:41 bribri1 node_exporter[46129]: ts=2026-06-05T20:51:41.201Z caller=collector.go:169 level=error msg>  
Jun 05 20:51:56 bribri1 node_exporter[46129]: ts=2026-06-05T20:51:56.225Z caller=collector.go:169 level=error msg>  
Jun 05 20:52:11 bribri1 node_exporter[46129]: ts=2026-06-05T20:52:11.178Z caller=collector.go:169 level=error msg>  
Jun 05 20:52:26 bribri1 node_exporter[46129]: ts=2026-06-05T20:52:26.201Z caller=collector.go:169 level=error msg>  
lines 1-20/20 (END)  
[4]+ Stopped sudo systemctl status node_exporter  
bribri@bribri1:~$ sudo systemctl status nginx  
● nginx.service - A high performance web server and a reverse proxy server  
Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)  
Active: active (running) since Fri 2026-06-05 11:12:45 UTC; 9h ago  
Docs: man:nginx(8)  
Main PID: 46916 (nginx)  
Tasks: 2 (limit: 4601)  
Memory: 1.6M (peak: 1.8M)  
CPU: 67ms  
CGroup: /system.slice/nginx.service  
└─46916 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"  
└─46917 "nginx: worker process"  
  
Jun 05 11:12:45 bribri1 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy se>  
Jun 05 11:12:45 bribri1 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy se>  
lines 1-14/14 (END)  
[5]+ Stopped sudo systemctl status nginx  
bribri@bribri1:~$ sudo systemctl status nginx_exporter  
× nginx_exporter.service - Ngix Prometheus Exporter
```

Conclusion

Ce TD m'a permis de mettre en place une solution de supervision complète basée sur Prometheus, Node Exporter et Grafana sur un serveur Ubuntu. J'ai appris à installer et configurer ces outils, à collecter des métriques systèmes (CPU, RAM, disque, réseau) et à les visualiser en temps réel via des dashboards Grafana.

J'ai également rencontré des difficultés comme la gestion des verrous apt et la configuration SSH sous VirtualBox, ce qui m'a permis de renforcer mes compétences en administration système.

Au final, la tâche est réussie et je suis désormais capable de déployer une stack de monitoring fonctionnelle sur un serveur Linux.